Auteur: Seda Jusjajeva
Studierichting: Informatica
Oefeningenreeks 3E nr. 4

$$V(t) = 5000 - 5000\left(1 - \frac{t}{40}\right)^2$$

$$V'(t) = -5000 \cdot 2 \cdot \left(1 - \frac{t}{40}\right) \cdot \left(\frac{-1}{40}\right)$$

$$= -10000 \cdot \left(\frac{-1}{40} + \frac{t}{1600}\right)$$

$$= 250 - \frac{25t}{4}$$

4.1:

$$V'(5) = 250 - \frac{25 \cdot 5}{4} = \frac{1000 - 125}{4} = \frac{875}{4}$$

De snelheid is $\frac{875}{4}$ l/min.

4.2:

$$V'(10) = 250 - \frac{25 \cdot 10}{4} = \frac{1000 - 250}{4} = \frac{375}{2}$$

De snelheid is $\frac{375}{2}$ l/min.

4.3:

$$V'(20) == 250 - \frac{25 \cdot 20}{4} = \frac{1000 - 500}{4} = 125$$

De snelheid is 125 l/min.

References